Ref. [33] — Why Gradient Clipping Accelerates Training

J. Zhang, T. He, S. Sra, A. Jadbabaie | ICLR 2020

1. Problem: why does clipping help optimization?

Classical proofs choose step sizes from a global gradient-Lipschitz constant: $\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\|$. Deep-network training can have strongly varying local smoothness, and large gradients often appear together with high curvature. Zhang et al. ask whether a different regularity assumption can explain why *normalizing or clipping updates* speeds up training.

2. The new smoothness model

A canonical form of the relaxed condition is

$$\|\nabla^2 f(x)\| \le L_0 + L_1\|\nabla f(x)\|,$$

where $L_0, L_1 \ge 0$. The important difference is that smoothness can increase with gradient magnitude; it need not be globally bounded by a constant independent of x. The paper motivates the condition with empirical measurements along neural-network trajectories.

3. Contrast the updates

Full gradient descent takes

$$x_{t+1} = x_t - \eta\nabla f(x_t).$$

For large $\|\nabla f(x_t)\|$, this may be a long step exactly when local curvature is large. A typical clipped direction is

$$\widetilde{g}_t = g_t \min\left\{1, \frac{c}{\|g_t\|}\right\}, \qquad x_{t+1} = x_t - \eta\widetilde{g}_t,$$

while normalized gradient descent uses $x_{t+1} = x_t - \eta g_t/\|g_t\|$ (with a convention at $g_t = 0$). Clipping bounds the update norm without necessarily changing small gradients.

4. First-principles reason for adaptivity

In the simple scalar teaching function $f(x) = e^x$, one has $f'(x) = f''(x) = e^x$. Curvature is proportional to gradient magnitude; no finite global L exists on all $\mathbb{R}$. For large x, a fixed GD step of size ηe^x can be enormous. A normalized step has length η, independent of e^x. This is an *illustration of the proposed mechanism*, not an experiment reported in the paper.

Main result. Under the relaxed smoothness model and the paper's other conditions, clipping and normalized-gradient methods admit faster convergence guarantees than fixed-stepsize GD; the authors give examples with arbitrarily large theoretical separation and test the explanation empirically. This is an adaptivity/first-order training-dynamics result, *not* a theorem guaranteeing escape from all saddles.

5. Limits and importance for the survey

The bound is a model assumption, not an established universal law of all neural-network objectives. Theoretical acceleration depends on the function class, choice of step/clipping parameters and stationarity target. ReLU non-differentiability and heavy-tailed noise require their own treatments. More recent adaptive-optimizer analyses borrow the motivation but need not share the same theorems.

Survey relevance (RQ2, RQ4). This paper is evidence that constant global smoothness is often an unrealistic lens. A clipped or normalized step responds to local gradient magnitude and curvature, which helps explain empirical robustness. Do not reinterpret its first-order rates as second-order or global guarantees.

6. Our notes

Clipping is not just a trick to prevent exploding gradients: it can be helpful because steps should shrink in regions where big gradients are accompanied by high curvature. The paper proves improvements under a relaxed smoothness condition; whether that condition holds for a given training setup still needs checking. **Original source:** https://sls.csail.mit.edu/publications/2020/TianxingHe_ICLR-2020.pdf.